

YUROU LIU

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EDUCATION

YALE UNIVERSITY

New Haven, CT

B.S. in Physics (Intensive)

Aug 2022 - May 2026 (anticipated)

B.S. in Computer Science

GPA: 3.99/4.00 Overall; 4.00/4.00 in Physics (Intensive)

Relevant Courses: Exoplanets, Many-Body Dynamics, Observational Astronomy ([incl. telescope proposal](#)), Cosmology, Research Methods in Astrophysics ([incl. telescope proposal](#)), Machine Learning, Classical Mechanics, Electromagnetism, Quantum Mechanics I & II, Statistical Mechanics, Advanced Physics Laboratory, Algorithms, Systems Programming, Computer Vision, Computer Graphics

RESEARCH EXPERIENCE

INDIANA UNIVERSITY BLOOMINGTON

Bloomington, IN

Alexander P. Hixon Summer Research Fellow, Department of Astronomy

Jun 2025 - Present

Mentored by Prof. Cristobal Petrovich and Dr. Hareesh Bhaskar

- Developed novel secular integration codes in C to study secular chaos in hierarchical “3+1” quadruple exoplanet systems (planet orbiting a star, perturbed by a nearby substellar/stellar companion and a fourth distant stellar companion)
- Created surfaces of section from rotating frame Hamiltonian of the “3+1” quadruple system to analytically probe parameter space of chaos
- Demonstrated that near-coplanar and circular “3+1” quadruple systems can excite planetary eccentricities to levels sufficient for high-eccentricity migration
- Manuscript submitted to AAS Journals and currently under review.

CALIFORNIA INSTITUTE OF TECHNOLOGY

Pasadena, CA

Summer Undergraduate Research Fellow, Department of Astronomy

Jun 2024 - Present

Mentored by Dr. Yapeng Zhang and supervised by Prof. Dimitri Mawet

- Performed atmospheric retrievals on CRIRES+ spectra of directly imaged gas giant 2M0249-0557 c and two benchmark brown dwarfs in the β Pictoris moving group
- Developed a nested sampling retrieval code using petitRADTRANS and PyMultiNest to constrain C/O ratio, metallicity, and $^{12}\text{CO}/^{13}\text{CO}$ isotopologue ratio
- Interpreted retrieved compositions in the context of formation pathways, supporting a star-like formation mechanism for 2M0249-0557 c
- Manuscript submitted to AAS Journals and currently under review.

YALE UNIVERSITY

New Haven, CT

Undergraduate Researcher, Department of Astronomy

Jun 2023 – Present

Mentored by Prof. Malena Rice and Dr. Tiger Lu

- Ran N-body simulations with REBOUND and REBOUNDx to model mirrored ZLK migration of two hot Jupiters in binary star systems
- Estimated occurrence rates of double hot Jupiters using the observed distribution of Gaia binaries hosting hot Jupiters
- Quantified the impact of mass and inclination asymmetries on double hot Jupiter formation
- Results published in the Astrophysical Journal; follow-up paper examining a case study of the WASP-94 system submitted to ApJL and currently under review

AWARDS AND FELLOWSHIPS

Alexander P. Hixon Fellowship, Yale University, 2025

Summer Undergraduate Research Fellowship, California Institute of Technology, 2024

First-Year Summer Research Fellowship in the Sciences & Engineering, Yale University, 2023

PUBLICATIONS

Liu, Y., Bhaskar, H. and Petrovich, C. [High-Eccentricity Tidal Migration Driven by Secular Chaos in Wide-Binary Systems](#). Submitted to AAS Journals (under review)

Liu, Y., Lu, T. and Rice, M. [Double Hot Jupiter Formation through Mirrored ZLK Migration in Binary Star Systems: The Case of WASP-94](#). Submitted to ApJL (under review)

Liu, Y., Zhang, Y., Xuan, J., Mawet, D., Snellen, I., Stolker, T., Kesseli, A. and Rice, M. [Chemistry and Isotope Ratios of Substellar Atmospheres in the \$\beta\$ Pictoris Young Moving Group](#). Submitted to AAS Journals (under review)

Liu, Y., Lu, T. and Rice, M., 2025. [The Formation of Double Hot Jupiter Systems through von Zeipel–Lidov–Kozai Migration](#). *ApJ* 986, 103

Lu, T., Tager, H., Hernandez, D.M., Rein, H., **Liu, Y.** and Rice, M., 2025. [Collisional Fragmentation Support in TRACE](#). *RNAAS* 9, 110

ORAL PRESENTATIONS (*INVITED)

Princeton Exoplanet Lunch*	Princeton, NJ, USA
<i>Gas Giants in Hierarchical Systems: from Dynamics to Atmospheres</i>	Sep 2025
Emerging Researchers in Exoplanet Science Symposium X	Princeton, NJ, USA
<i>Chemistry and Isotope Ratios of Substellar Atmospheres in the β Pic Moving Group</i>	Jun 2025
246 th American Astronomical Society Meeting	Anchorage, AK, USA
<i>Chemistry and Isotope Ratios of Substellar Atmospheres in the β Pic Moving Group</i>	Jun 2025
Boston Area Planetary Science Meeting	Boston, MA, USA
<i>Chemistry and Isotope Ratios of Substellar Atmospheres in the β Pic Moving Group</i>	May 2025
Caltech Student-Faculty Programs Seminar Day	Pasadena, CA, USA
<i>Chemistry and Isotope Ratios of the Directly Imaged Gas Giant 2MJ0249-0557c</i>	May 2025
The 55 th Meeting of the Division of Dynamical Astronomy	Toronto, Canada
<i>The Formation of Double Hot Jupiter Systems Through ZLK Migration</i>	May 2024
Yale Astronomy Department Summer Research Symposium	New Haven, CT, USA
<i>The Dynamical History of the Two Hot Jupiters in the WASP-94 Binary Star System</i>	Jul, 2023

POSTERS

New York Area Exoplanets Meeting	New York, NY, USA
<i>Chemistry and Isotope Ratios of Substellar Atmospheres in the β Pic Moving Group</i>	May 2025

OBSERVING EXPERIENCE

Palomar Observatory (DBSP): 1 night
Spectroscopy of a Gas Giant Exoplanet in Optical Wavelength

TECHNICAL SKILLS

Languages: Python, C/C++, D, Rust, R
Scientific/Graphics: REBOUND, REBOUNDx, KozaiPy, petitRADTRANS, PyMultiNest, TensorFlow, PyTorch, matplotlib, NumPy, SciPy, Astropy, pandas, GSL, Mathematica, OpenGL, SDL
Tools: Git, Bash, LaTeX, SQL, Slurm

PROJECTS

Generate Fake Transits

- Trained a conditional variational autoencoder on TESS data to generate synthetic exoplanet transits based on input parameters.

L4 and L5 Lagrange Points Visualization

- Visualized particle dynamics near Lagrange points in a star-planet system using OpenGL instanced rendering.

More projects at: yurouninaliu.github.io/projects

TEACHING EXPERIENCE

YALE UNIVERSITY

Undergraduate Learning Assistant, Department of Physics

PHYS 261: Intensive Introductory Physics II

PHYS 260: Intensive Introductory Physics I

New Haven, CT

Sep 2023 – May 2024

OUTREACH TALKS

Yale Undergraduate Kick-off Event

Hot Jupiter Formation through Secular Chaos in “3+1” 4-body Systems

New Haven, CT, USA

Sep 2025

SCIENCE COMMUNICATION AND OUTREACH

Contributor and layout editor, Yale Scientific Magazine

- Created original artwork and designed print layouts
- Provided individualized feedback to contributors
- Improved layout designs for publication
- Led layout workshops to train new designers
- Volunteered at the Resonance Conference to facilitate hands-on STEM classes for local high school students

Workshop leader, Yale Guild of Book Makers

- Taught hands-on workshops in bookbinding

MEDIA

Yale News, “[New study offers a double dose of ‘hot Jupiters’](#)”

Jun 2025

StarXiv, “[Episode 11: Whirling planes, wandering black holes and alien supernovae](#)”

May 2025